

Ashley Daniel Ulumben

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SUMMARY

Data Scientist with an M.S. in Computer Science and experience using Python, SQL, PostgreSQL, and machine learning to build analysis-ready datasets, models, dashboards, and reports. Skilled in ETL, data validation, feature engineering, regression/classification modeling, model evaluation, and translating results into actionable insights for technical and non-technical stakeholders.

TECHNICAL SKILLS

Programming & Analytics: Python (Pandas, NumPy, Scikit-learn), SQL, PostgreSQL, TypeScript, React

Machine Learning: Regression, classification, Random Forest, decision trees, feature engineering, ROC-AUC, PR-AUC, F1-score, confusion matrices

Data Pipelines & Visualization: ETL, API/JSON processing, relational schema design, data validation, Power BI, Matplotlib, Seaborn, Recharts

Cloud & Tools: AWS S3, CloudFront, Amplify, Supabase, Git/GitHub, Linux fundamentals, AWS Certified Cloud Practitioner

PROFESSIONAL EXPERIENCE

The University of Alabama - Tuscaloosa, AL

Research Assistant

Aug 2025 - Present

- Preprocess GIS, elevation, rainfall, and population datasets with Python to create validated, analysis-ready inputs for flood scenario modeling.
- Build data workflows that integrate real-world geospatial data into Unity/C# simulations and support decision-based flood resilience analysis.

Auburn University - Auburn, AL

Research Assistant

Jun 2025 - Present

- Build a full-stack analytics application using React, TypeScript, PostgreSQL, and Supabase to support data entry, metric tracking, and dashboard reporting.
- Design relational schemas and SQL queries to extract, transform, and validate hatchery datasets across multiple operational metrics.
- Develop dashboard views and deploy the application using AWS S3, CloudFront, and Amplify with Git-based workflows for shared development.

Genius Minds LLC - Boston, MA

Business Analyst

Oct 2024 - Jun 2025

- Performed ETL on Web API JSON data using Python and SQL, transforming semi-structured records into PostgreSQL tables for analytics.
- Designed and maintained a 7-table relational database to support recurring reporting, dimensional analysis, and client-facing insights.
- Wrote SQL queries to analyze datasets, validate data quality, and produce scheduled reports for business decision-making.

Education Policy Center - Tuscaloosa, AL

Graduate Research Assistant

Jan 2024 - May 2024

- Built a Random Forest Regression model in Python to analyze multi-year U.S. education datasets and predict enrollment, diversity, and financial-aid trends.
- Prepared and validated datasets, engineered analytical variables, and visualized model outputs with Matplotlib and Seaborn to communicate insights.

Alabama Transportation Institute - Tuscaloosa, AL

Graduate Student Assistant

Aug 2023 - Dec 2023

- Analyzed transportation route statistics using Python (Pandas) and Power BI to identify regions with low transit access and reporting gaps.
- Developed Python web-scraping scripts to collect rural county transportation metrics and improve repeatable data collection workflows.

PROJECTS

Public Health Risk Prediction and Model Evaluation | Python, SQL, Scikit-learn, XGBoost, Matplotlib

- Developed classification workflows using historical outbreak and state-level datasets to predict high-burden outcomes and compare model performance.
- Engineered lag, rolling-average, growth-rate, and weather features; evaluated models using ROC-AUC, PR-AUC, recall, F1-score, and confusion matrices.

Backend Web API Basketball Database Engineering | Python, SQL, PostgreSQL

- Designed a PostgreSQL database and built a Python ETL pipeline for loading, validating, and querying JSON-based game data.

EDUCATION

The University of Alabama - Tuscaloosa, AL

M.S. Computer Science

Aug 2022 - May 2024

University of Mumbai - Mumbai, India

B.S. Information Technology

Jul 2016 - May 2019